Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

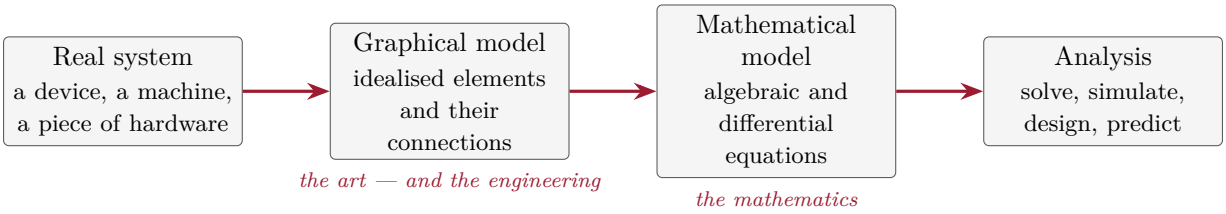
Lecture 1: Basic Concepts

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Colour code used in these notes and in the slides			
charge q	current i	voltage v	power p, energy w

1. Why circuits? From the world to a model

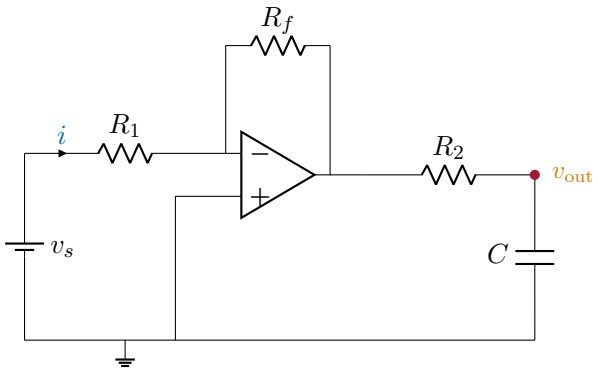
Circuit theory is not about wires. It is a way of *modelling* physical systems, and you will use the same thinking in control, signal processing, power, robotics and electronics for the rest of your career. The pipeline is always the same:



The hard step is the first arrow. A real resistor has inductance, a temperature coefficient, a power rating and a size; we throw all of that away and keep one number, R. Whether that is brilliant or foolish depends on what you ask of the model. Calling that step "the art" makes it sound decorative, as though the equations were the real work. The opposite is true: the mathematics can be looked up, while deciding what to keep and what to discard *is* the engineering.

> *"All models are wrong, but some are useful."*
>
> George E. P. Box, *Science and Statistics*, 1976

Every element you will meet in EE201 is wrong in some measurable way. The question is never "is this model true?" but "is it useful here?".



A circuit you will be able to analyse by the end of the semester: a source, resistors, an operational amplifier and a capacitor — one element from each part of the course. All of EE201 is about what each element does on its own, and what the wires force them to share.

The same pipeline, other domains

The elements change, the structure does not. Every such domain has one *across* quantity and one *through* quantity:

Domain	"Across" quantity	"Through" quantity
electrical	electric potential	current
hydraulic	pressure	volumetric flow rate
thermal	temperature	heat flow
mechanical	velocity	force

An across quantity is always a *difference* between two points, and in every domain the word "difference" is dropped. Sometimes for convenience: "the voltage across R_2" names no terminals, yet nobody is confused. More often because a *reference* was agreed on long ago — altitude from sea level, gauge pressure from atmospheric pressure (a flat tyre reads zero, not 101 kPa), temperature from absolute zero. A circuit has no universal reference, so we choose one and call it *ground*.
A through quantity is the opposite: measured *at* one place — a cross-section of a wire, a pipe, a shaft — it needs a direction, not a second point, and no reference either. Zero current means the same to everyone; zero volts is somebody's decision.
All these domains obey the same conservation and connection laws, which is why circuit theory is taught across engineering.

In EE201 we work with the *lumped circuit* model, made precise at the end of this lecture. The sections before it build its vocabulary: charge, current, voltage, ground, power, and a sign convention that keeps the bookkeeping honest.

2. Charge

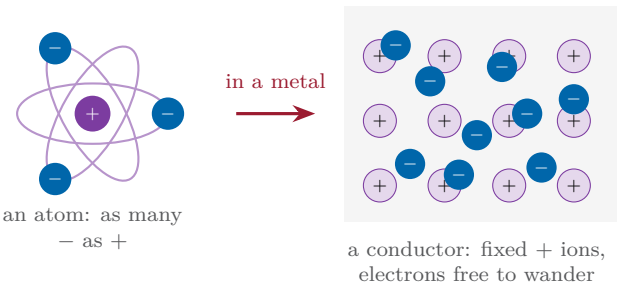
Electric charge

Charge is the physical property of matter that makes it experience a force in an electromagnetic field. It is denoted by q (or Q for a constant charge) and measured in **coulombs (C)**.

Charge is quantised: it comes in integer multiples of the elementary charge

$$e = 1.602 \times 10^{-19}\,\mathrm{C}, \qquad \text{so} \qquad 1\,\mathrm{C} \approx 6.24 \times 10^{18} \text{ electron charges.}$$

An electron carries $-e$ and a proton $+e$. In an isolated atom the two balance exactly, so the atom is neutral.



an atom: as many − as +

a conductor: fixed + ions, electrons free to wander

Two facts about charge do all the work in this course:

1. **Charge is conserved.** It is neither created nor destroyed; it only moves. When we write Kirchhoff's current law in Lecture 2, we are writing conservation of charge.

2. **A conductor is already full of charge, and is still neutral.** In a metal, roughly one electron per atom breaks loose, leaving a fixed positive ion behind. One cubic centimetre of copper contains about 8.5×10^{22} of these — about 13 600 C of *mobile* charge — yet its net charge is essentially zero, because every loose electron is matched by the ion it left.

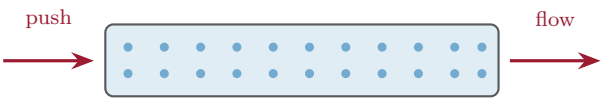
What "free" does and does not mean

These are traditionally called **free electrons**, the term you will meet in your solid-state and semiconductor courses. But *free* here means free to move **within** the metal — not free to leave it. The electron wanders through the lattice; it does not escape, and the ion it came from stays put. That is why a piece of copper can hold 13 600 C of movable charge per cubic centimetre and still be neutral. When the two ideas seem to contradict each other, this word is the trouble; in these notes the adjective is **mobile**.

The second fact is easy to misread. A current in a wire is not charge being *poured into* the wire: the charge is already there, and a current is that charge *drifting*. The wire stays neutral while it drifts. This is the first assumption of the lumped model below, and what makes Kirchhoff's current law true.

2.1. A pipe that is already full

The hydraulic picture is the right one, provided you start it correctly: a wire is not an empty pipe waiting to be filled, but a pipe that is *already full*.



Raise the pressure at one end and the far end flows *at once* — long before any drop of water has travelled the length of the pipe.

Read across from the picture and most of this lecture is already in it:

pipe already full of water	**wire already full of mobile charge**
pressure difference between the ends	**voltage** across the element
rate of flow past a point	**current** through it
the far end responds immediately	the lamp lights the instant you switch it on
water circulates, it is not consumed	charge is not consumed either
flow needs a closed loop	so does a circuit

The last two rows head off the two commonest wrong pictures: that a lamp *uses up* the current passing through it, and that charge can be delivered down a single wire. Neither survives the full pipe. Whatever flows in at one end flows out at the other, and it has to come back around.

3. Current

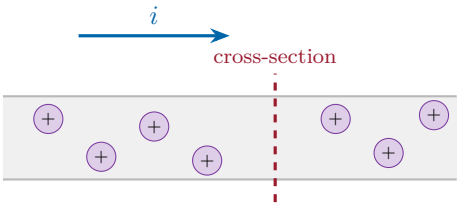
If charge sits still, nothing interesting happens. Circuits are about charge *in motion*.

Electric current

Current is the time rate of flow of charge through a cross-section:

$$\boxed{i(t) = \frac{dq(t)}{dt}} \quad \Longleftrightarrow \quad q(t) = q(t_0) + \int_{t_0}^{t} i(\tau)\, d\tau.$$

The unit is the **ampere (A)**, and $1\,\mathrm{A} = 1\,\mathrm{C/s}$.



$i = dq/dt$: charge crossing the dashed line per second

Example 1.1 — a coulomb, and the speed of an electron

(a) How much charge does a phone battery hold? A 4000 mAh cell:

$$Q = 4\,\text{A} \times 3600\,\text{s} = 14\,400\,\text{C}, \qquad N = \frac{14\,400}{1.602 \times 10^{-19}} \approx 9 \times 10^{22} \text{ electrons.}$$

The rating is in ampere-hours — a charge written as a current times a time. Charge comes in whole electrons, but with 10^{22} of them we can treat $q(t)$ as a smooth function, which is what lets us write dq/dt at all.

(b) How fast do the electrons actually move? Take 1 A flowing in an ordinary 1 mm^2 copper wire. Each electron drifts along the wire at roughly

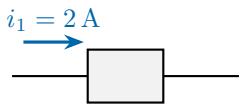
$$0.07 \text{ mm/s} \qquad \text{— about one metre in four hours.}$$

Yet the lamp lights the instant you flip the switch. The wire is already full, so you never wait for an electron to travel from the switch to the lamp: every electron along it starts moving together, almost the moment you close the circuit.

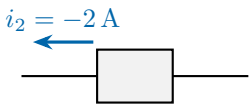
(For the curious: the drift speed follows from $v = i/(neA)$, with $n \approx 8.5 \times 10^{28}$ m^{-3} the density of mobile electrons in copper. Not needed in EE201 — it is here so the number is not magic.)

3.1. Reference direction: the arrow is a choice, not a fact

A current value is meaningless without an arrow. The arrow declares *which way we call positive*; the sign of the number then tells us which way the charge actually moves.



charge flows to the right



exactly the same physical situation

Both pictures describe the same current. You will often assign arrows before you know the answer; a negative result is not a mistake, it is the model telling you the charge goes the other way.

Conventional current

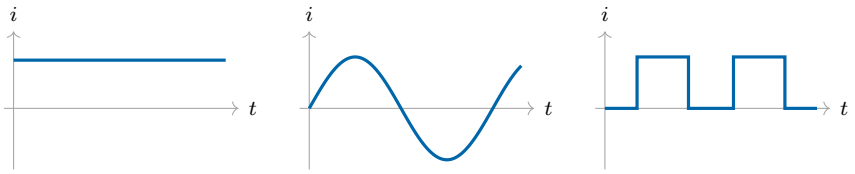
By convention, the current arrow points the way *positive* charge would move. In a copper wire the carriers are electrons, so they move *against* the arrow.

No equation in this course changes because of it. A current counts only the *net rate at which charge crosses a section*, and an electron moving left carries as much charge to the right as a proton moving right. The model never asks what the carriers are — and it had better not, since in a semiconductor they are electrons *and* holes, and in a battery ions.

3.2. Notation

The usual convention: lowercase letters for quantities that may vary with time, uppercase for constants.

$$i(t),\ v(t),\ q(t),\ p(t) \quad \text{(general)} \qquad\qquad I,\ V,\ Q,\ P \quad \text{(constant, “DC”)}$$



Example 1.2 — charging a phone

In the last part of a charge cycle the battery is topped up: the charge *stored in the battery* approaches its full value as

$$q(t) = Q_f\left(1 - e^{-t/\tau}\right), \qquad Q_f = 3600\,\text{C}\ (= 1000\ \text{mAh}), \qquad \tau = 20\ \text{min} = 1200\,\text{s}.$$

The charging current is the rate at which that charge accumulates:

$$i(t) = \frac{dq}{dt} = \frac{Q_f}{\tau}\,e^{-t/\tau} = \frac{3600}{1200}\,e^{-t/\tau} = 3\,e^{-t/\tau}\ \text{A}.$$

The current starts at 3 A and decays: after 20 minutes it is $3/e \approx 1.1$ A, after an hour about 0.15 A. The fuller the battery, the more slowly it fills — which is why the last ten per cent takes so long.

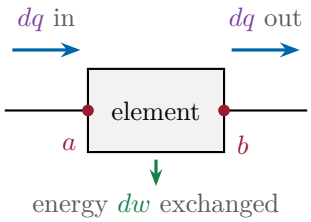
With a *constant* current the charge accumulates linearly: $I = 3$ A flowing for 8 hours transports $Q = I\,\Delta t = 3 \times 8 \times 3600 = 86\,400$ C.

Charge is what accumulates; current is how fast it accumulates. Every integral–derivative pair in this course has that same flavour.

4. Voltage

Current tells us charge is moving, but not *why* or what the motion costs. Charge does not wander through a circuit on its own: carrying it through an element exchanges energy, and voltage is the bookkeeping of that exchange.

Let a small charge dq enter an element at terminal a and leave at b, exchanging energy dw on the way.



Voltage (potential difference)

The **voltage** v_{ab} between two points a and b is the energy exchanged *per unit of charge* carried from a to b:

$$\boxed{v_{ab} = \frac{dw}{dq}}$$

The unit is the **volt (V)**, and $1\,\text{V} = 1\,\text{J/C}$. A positive v_{ab} means the charge *gives up* energy to the element on its way from a to b.

Four remarks, and they matter more than the formula.

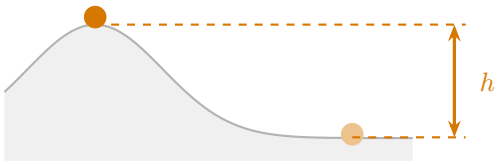
1. **It is a ratio.** Carry twice the charge and twice the energy is exchanged, so v_{ab} is the same. Voltage describes the *pair of terminals*, not the charge that passed between them.

2. **It does not need a current.** A 1.5 V battery in a drawer has 1.5 V across its terminals with $i = 0$, just as a closed tap still has pressure behind it — and it still has 1.5 V across it while driving 2 A through a lamp. Voltage is defined, measurable and computable in both cases: it is a readiness to exchange energy per unit charge, and says nothing by itself about whether charge is flowing. Voltage and current are separate quantities; it takes an *element* to relate them, which is what Lecture 3 is about.
3. **It is always between two points, and the order matters:** $v_{ab} = -v_{ba}$. "The voltage at node a" means nothing until a reference point is chosen — which is why the older name *potential difference* is the more honest one.
4. **In a lumped circuit it depends only on the two endpoints, not on the path between them.** Carry the charge from a to b by any route and the energy per unit charge is the same, so "the voltage between a and b" is a single, definite number. This is the second lumped-model assumption in disguise; applied around a closed path, it is Kirchhoff's voltage law.

One thing comes for free: with $dq = i\,dt$, the energy exchanged per unit time is $dw/dt = v\,i$ — the power, which needs no separate definition.

4.1. Voltage as an electrical height

The most useful picture is gravitational. Lifting a mass m through a height h costs mgh; the energy *per unit mass*, gh, depends only on the height difference. A heavier ball gains more energy, but energy per kilogram is the same — so gh, like voltage, belongs to the two places, not to the object carried between them.

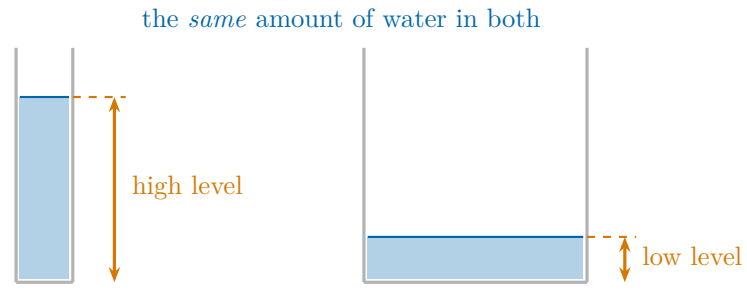


energy per unit *mass* to lift through h: $g\,h$

The picture anticipates two later ideas. "Sea level" is a choice, and moving it changes no height difference — that is ground. And the energy spent climbing a hill does not depend on the route — that is remark 4, and Kirchhoff's voltage law.

4.2. Charge is the amount, voltage is the level

A trap that catches almost everybody: charge starts to *feel* like energy. It is not. The way to see why is to separate the *amount* of something from its *level*.



Pour one litre into a narrow tube and the level jumps; pour the same litre into a wide tank and it barely moves. Same amount, different level, because the containers have different cross-sections. Electrically, the "cross-section" of a pair of conductors is its **capacitance** C, and the relation is the same:

$$q = C\,v.$$

The *shape of the circuit* plays the part of the shape of the tank: the same charge moved between a different pair of points gives a different voltage, because the geometry — and therefore C — is different. Capacitance is defined properly in Lecture 4; for now it is enough that the link between *how much charge* and *what voltage* is a property of the circuit, not a universal constant.

Why charge cannot be energy in disguise

Charge the pair of conductors up to q. The energy you had to supply is

$$W = \int_0^q v\,dq' = \int_0^q \frac{q'}{C}\,dq' = \frac{q^2}{2C} = \tfrac{1}{2}Cv^2.$$

Two things follow. First, energy grows with the *square* of charge — if charge were energy under another name, the relation would be linear. Second, the *first* coulomb costs almost nothing, because the level is near zero, while the *last* is expensive, because it must be lifted to the full height. The price of one coulomb *is* the voltage, and the voltage changes as you fill.

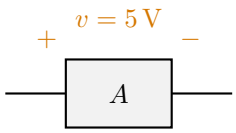
The sharpest statement of the difference is about a complete circuit:

Around a loop, the charge that leaves the source comes back to it. The energy does not.

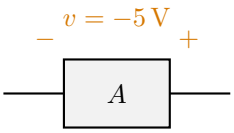
Charge *circulates*: every coulomb entering a resistor leaves it, none "used up". Energy flows one way — out of the source, into heat, light or motion, never back. The copper above makes the same point: one cubic centimetre holds some 13 600 C of mobile charge and stores no energy at all. Charge is everywhere; what carries energy is the *difference in level*.

4.3. Reference polarity

Exactly as for current, we *can* mark a reference polarity with + and − before we know the answer. The two pictures below describe the same element in the same state.



left terminal is 5 V above the right one



same element, *same* state

4.4. The first element: the resistor

So far we have defined only *variables*: charge, current, voltage. A circuit also needs *elements*, each imposing a rule between the voltage across it and the current through it. That rule is the element's **branch relation**; the whole family arrives in Lecture 3. One is worth meeting now, because the analogies below need it.

The resistor and Ohm's law

A **resistor** is a two-terminal element whose voltage and current obey, at every instant, the algebraic relation

$$\boxed{v = Ri}$$

where R is the **resistance**, measured in **ohms** (Ω), with $1\,\Omega = 1\,\mathrm{V}/\,\mathrm{A}$.



Two features of this relation matter more than the formula.

- **It is algebraic, and has no memory.** The current now is fixed by the voltage now — no derivatives, no integrals, nothing about the past. Such elements are called *resistive* (or *static*), and the first four topics of the course live entirely among them. Capacitors and inductors, which *do* remember, arrive in Lectures 4 and 5 and change the problem completely.
- **The resistance comes from geometry and material.** For a uniform bar of length L and cross-section A made of a material of resistivity ρ,

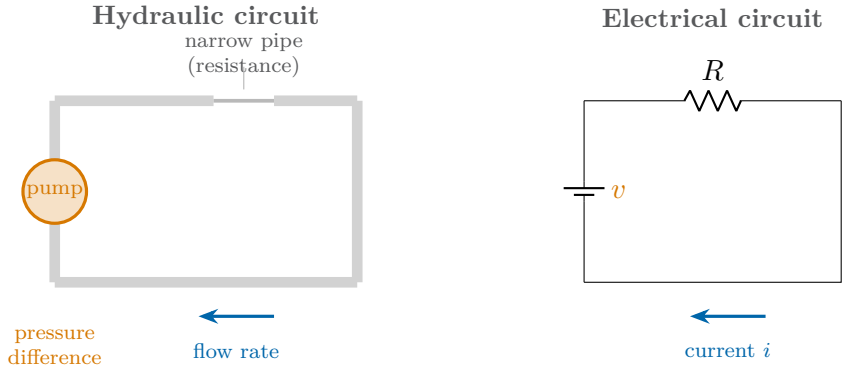
$$R = \rho \frac{L}{A}.$$

  Long and thin means large R; short and fat means small R — exactly the behaviour of a pipe carrying water, which is why the next subsection's analogy works so well. (Copper has $\rho \approx 1.7 \times 10^{-8}\ \Omega \cdot \mathrm{m}$, which is why we make wires out of it and assume they have no resistance at all.)

The sign convention that accompanies $v = Ri$, and the reason a resistor can only *absorb* energy, come once power is defined.

4.5. The hydraulic analogy

Most people have better intuition for water than for electrons, and the analogy is close enough to be useful — as long as you remember it is an analogy.



Hydraulic	Electrical	Comment
pressure (difference)	voltage v	the "across" quantity: a difference between two points
volumetric flow rate	current i	the "through" quantity: needs a direction
pipe / tap restriction	resistance R	opposes flow, dissipates energy
pump	voltage source	supplies energy
water in the pipes	charge q	conserved: it does not disappear

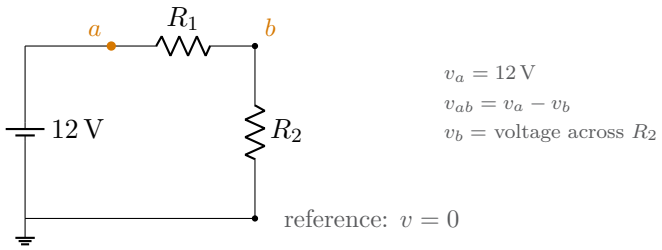
5. Ground and node voltages

Since only *differences* of potential have physical meaning, we may declare one point of the circuit to be zero. That point is the **reference node**, or **ground**.

Ground (reference node)

A node chosen, by us, to have potential zero. The "voltage at node a", written v_a, then means $v_{a,\mathrm{ground}}$, and every two-point voltage follows from

$$\boxed{v_{ab} = v_a - v_b}$$



Choosing a different ground shifts *every* node voltage by the same constant and changes *no* two-point voltage, current or power. The choice is convenience: pick the node the most elements touch, and your equations get shorter.

Three different things are called "ground"

Earth ground is a physical connection to the planet, used for safety. **Chassis ground** is the metal enclosure of an instrument. **Signal ground** is the analysis reference we have just defined. In EE201 we mean the third one; in the laboratory, confusing them is how you blow a fuse.

6. Power and energy

Power and energy

Power is the rate at which energy is transferred:

$$\boxed{p(t) = \frac{dw}{dt} = \frac{dw}{dq}\cdot\frac{dq}{dt} = v(t)\,i(t)}$$

The unit is the **watt (W)**, $1\,\mathrm{W} = 1\,\mathrm{J/s}$. The energy transferred between t_0 and t is

$$w(t) = \int_{t_0}^{t} p(\tau)\,d\tau = \int_{t_0}^{t} v(\tau)\,i(\tau)\,d\tau.$$

$p = vi$ is not a separate definition: it follows from voltage (energy per charge) and current (charge per time). That is the whole point of defining them the way we did.

For the one element we have met, Ohm's law gives

$$p = vi = i^2R = \frac{v^2}{R},$$

which for $R > 0$ is never negative, whatever the signs of v and i. A resistor can only *absorb* energy — it has no way of giving any back — and that energy leaves the circuit as heat.

Energy bills are written in kilowatt-hours: $1\ \mathrm{kWh} = 10^3 \times 3600 = 3.6 \times 10^6\,\mathrm{J}$.

A units check: a 12 V battery supplying a constant 2 A for 5 minutes delivers

$$P = VI = 12 \times 2 = 24\,\mathrm{W}, \qquad W = P\,\Delta t = 24 \times 300 = 7200\,\mathrm{J} = 7.2\ \mathrm{kJ} = 2 \times 10^{-3}\ \mathrm{kWh}.$$

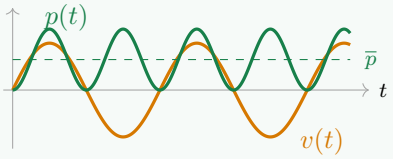
Example 1.3 — power that changes in time

A resistive element carries

$$v(t) = 10\cos(\omega t)\,\text{V}, \qquad i(t) = 2\cos(\omega t)\,\text{A}, \qquad \omega = 2\pi \cdot 50\ \text{rad/s}.$$

The instantaneous power is

$$p(t) = v(t)\,i(t) = 20\cos^2(\omega t) = 10(1 + \cos 2\omega t)\,\text{W}.$$



- $p(t) \geq 0$ always: a resistor never returns energy.
- The power *oscillates* at twice the frequency of the voltage, about an average of 10 W.
- The *peak*, 20 W, is twice the average. An element must survive the peak; how hot it gets is set by the average.

The energy absorbed over one period $T = 1/50$ s is

$$w = \int_0^T p(t)\,dt = 10\,T + \underbrace{\int_0^T 10\cos 2\omega t\,dt}_{=0} = \frac{10}{50} = 0.2\,\text{J}.$$

The average power is what a wattmeter shows and what heats the element; the 100 Hz oscillation is what makes a cheap transformer hum.

7. The passive sign convention

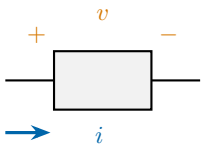
We have $p = vi$, but v and i each carry a reference that *we* chose. Which sign of p means "this element is consuming energy"? A convention settles it.

Passive sign convention (PSC)

Assign the current reference arrow so that it **enters the terminal marked** $+$. With that choice,

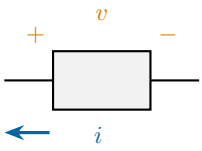
$$\boxed{p = vi \;=\; \text{power } \textit{absorbed} \text{ by the element}}$$

so $p > 0$ means the element absorbs (consumes) energy, and $p < 0$ means it delivers (supplies) energy to the rest of the circuit.



$v > 0$, i into $+$

$p = vi$, **element absorbs**



$v > 0$, i out of $+$

$p = vi$, **element delivers**

If the picture does *not* satisfy the PSC — the arrow leaves the $+$ terminal — no second formula is needed. Flip one reference, change that variable's sign, and use $p = vi$:

$$p_{\text{absorbed}} = v\,(-i) = -vi.$$

7.1. Conservation of power

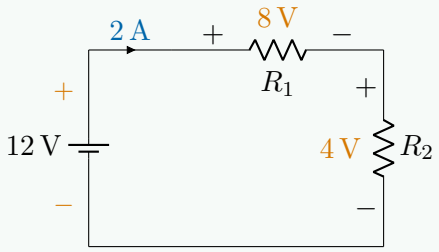
Energy is conserved, so in *any* circuit, at *every* instant, the absorbed powers of all elements sum to zero:

$$\boxed{\sum_{k=1}^{N} p_k(t) = 0}$$

Elements with $p_k < 0$ (usually sources) supply exactly what those with $p_k > 0$ (usually resistors) consume. This is Tellegen's theorem: if your powers do not cancel, something is wrong.

Example 1.4 — a power balance

In the circuit below the source drives 2 A around the loop and the two resistors drop 8 V and 4 V.



With the PSC applied to each element:

$$p_{R_1} = 8 \times 2 = +16\,\text{W}, \qquad p_{R_2} = 4 \times 2 = +8\,\text{W}, \qquad p_{\text{source}} = 12 \times (-2) = -24\,\text{W}.$$

The source current leaves its + terminal, hence the −2 A; the sum is $16 + 8 - 24 = 0$, as it must be. The source *delivers* 24 W and the resistors *absorb* it.

8. Electric circuits and the lumped model

Electric circuit

An **electric circuit** is an interconnection of electrical elements through which charge can flow along closed paths.

Everything we do in EE201 rests on one further modelling assumption, worth stating precisely rather than assuming silently.

The lumped circuit model

A circuit is **lumped** when three things may be assumed.

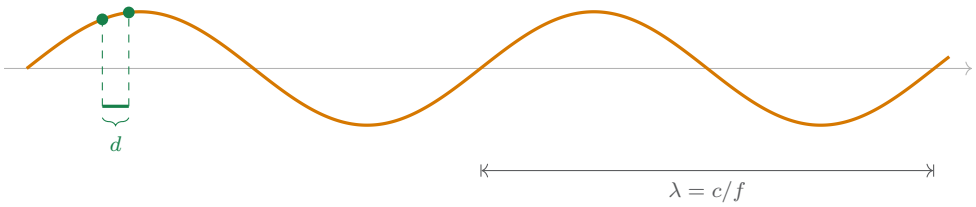
(a) Nothing piles up. At every instant, the current leaving an element equals the current entering it.
(b) Nothing leaks out. No changing *magnetic* field passes through the loops formed by your wiring, and no charge on one element pulls on the charge in another across the space between them. The fields belong to the elements, not to the space between them.
(c) Nothing is late. A signal of frequency f travelling at the speed of light c advances a distance $\lambda = c/f$ in one period; λ is its **wavelength**. The circuit, of size d, must be much smaller than that:

$$\boxed{d \ll \lambda = \frac{c}{f}}, \qquad c \approx 3 \times 10^8 \text{ m/s}.$$

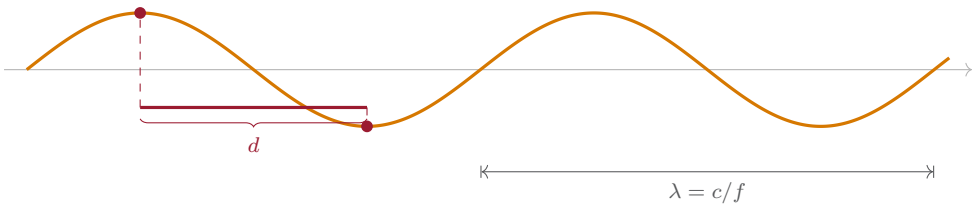
Read (c) in terms of times if you prefer: crossing the circuit takes d/c, the signal changes in $1/f$, and we assume the first is negligible next to the second. Either way, a change at one point reaches everywhere else before anything else has had time to happen.

When the three hold, a circuit is described by *ordinary* differential equations in time. When they do not, the same physics needs Maxwell's equations and partial differential equations in space and time — what the electromagnetic field theory courses are for.

LUMPED: $d \ll \lambda$ both ends of the circuit sit at practically the same point of the wave



NOT LUMPED: $d \sim \lambda$ one end is at the crest while the other is at the trough



a *snapshot* along the circuit at one instant — the horizontal axis is distance, not time

The picture above is the one to keep. It shows the signal *frozen in space*: the horizontal axis runs along the circuit, and the curve is the signal's value at each place at one instant. If the circuit occupies a tiny fraction of a wavelength, every point reads the same number and we may speak of "the" voltage of a node. If it occupies an appreciable fraction, different parts hold different values at the same moment, and a single node voltage no longer exists.

Each of the three, translated:

- **(a) does not forbid charge from moving *inside* an element.** A capacitor is the obvious case: it puts $+q$ on one plate and $-q$ on the other, so charge does separate inside it. (a) says the element as a *whole* never gains or loses charge — the same current $i = dq/dt$ that enters one terminal leaves the other. A wire, likewise, is a pipe and not a bucket. Without this, "the current through an element" would not be well defined, and Kirchhoff's current law would be false.
- **(b) says the loops in your diagram are not antennas and not transformers.** You drew a rectangle of wire; a changing *magnetic* field through it would induce a voltage, and the voltage between two nodes would depend on how you routed the wire. Assumption (b) rules that out, which makes "the voltage between a and b" independent of the path — and Kirchhoff's voltage law true.
- **(c) says everything happens everywhere at once.** A signal crosses a 30 cm board in about a nanosecond, while a 50 Hz mains waveform takes milliseconds to change appreciably — a factor of a million. That delay is not worth modelling, so one time variable t serves the whole circuit, with no dependence on position.

LUMPED

mains, $f = 50$ Hz $\Rightarrow \lambda = 6000$ km
a 30 cm board: $d/\lambda \sim 10^{-7}$
— circuit theory applies

NOT LUMPED

Wi-Fi, $f = 2.4$ GHz $\Rightarrow \lambda = 12.5$ cm
a 3 cm trace: $d/\lambda \sim 1/4$
— you need field theory

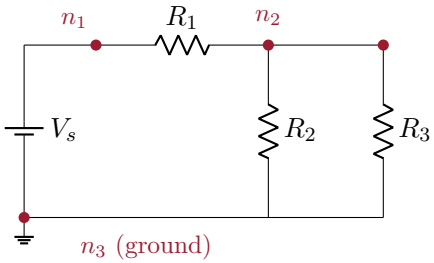
The same hardware can be lumped for one signal and distributed for another. A mobile phone motherboard is a lumped circuit for its battery and a transmission-line problem for its antenna. The model is chosen by the question, not by the object.

9. Terminals, nodes, branches

The last piece of vocabulary describes how elements are connected — the topology.

Terminal, node, branch, loop

- A **terminal** is a point at which an element can be connected to others. A resistor, capacitor or inductor has two terminals; a transistor has three; an op-amp has five or more.
- A **node** is a point of connection between two or more elements. Everything joined by ideal (zero-resistance) wire is *one and the same node*, however long the wire is drawn.
- A **branch** is a single element between two nodes.
- A **loop** is a closed path through the circuit that does not pass through any node more than once.

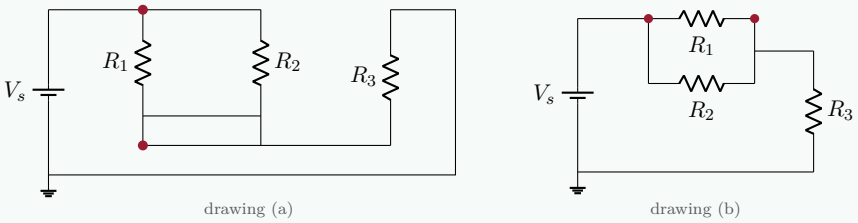


In the figure, n_2 is a single node even though three elements meet there through a stretch of wire; the wire itself is not a branch. Counting: 3 nodes, 4 branches (V_s, R_1, R_2, R_3) and 2 independent loops, consistent with the relation you will meet in Lecture 2,

$$\#\text{branches} = \#\text{loops} + \#\text{nodes} - 1 \qquad \Longrightarrow \qquad 4 = 2 + 3 - 1.$$

Example 1.5 — the same circuit, drawn twice

The two drawings below look different but are the *same* circuit: in both, R_1 and R_2 share both terminals, and that pair is in series with R_3 and the source.



Drawing (b) makes it obvious that R_1 and R_2 are in parallel; drawing (a) hides it behind a detour of wire. Only the placement of the symbols changed — and that is the only skill being tested when a problem looks impossible at first sight.

Drawing is not topology

Two circuits that look completely different on paper can be the same circuit. What matters is which terminals share a node, not where you placed the symbols. So when a problem looks unfamiliar, redraw it: put every element on a given node so that it visibly touches the *same point or the same line*, and straighten the detours of wire. Series and parallel connections usually become obvious, and half the difficulty goes with them.

Summary

Quantity	Symbol	Defining relation	Unit
charge	q	—	coulomb, C
current	i	$i = dq/dt$	ampere, A = C/s
voltage	v	$v = dw/dq$	volt, V = J/C
power	p	$p = dw/dt = vi$	watt, W = J/s
energy	w	$w = \int p\,dt$	joule, J

- Current needs a **reference arrow**; voltage needs a **reference polarity**. Both are choices. A negative answer is information, not an error.
- With the **passive sign convention** (current into the + terminal), $p = vi$ is the absorbed power, and $\sum_k p_k = 0$ over the whole circuit.
- Voltage is an **across** quantity, current is a **through** quantity; ground is the node we *choose* to call zero.
- The **lumped** assumption ($d \ll \lambda$, no charge accumulation, no stray flux) is what lets us use ordinary differential equations for the rest of the course.

Next lecture

Interconnection laws — Kirchhoff's current and voltage laws — and the branch relations of the basic elements. Everything from Lecture 1 is used there without further comment.